Royal Devon University Healthcare NHS Foundation Trust

Patient mobilisation with an endotracheal tube and vasopressors Northern Services	
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Note: This document has been assessed for any equality, diversity or human rights implications			

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		Consultation with relevant groups and governance groups	*Band 6/7/8 respiratory physiotherapists Consultation April / May 2023* *Judith Summers, specialist band 7 ICU nurse 2/5/23* *ICU senior team Governance meeting 26/5/23* *AHP team leaders / Helen Cooke 13/6/23* *Shared with Andrew Lockwood – Exeter ICU physiotherapy lead 20/06/23*

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KEY POINTS OF THIS PROCEDURAL DOCUMENT:

Guidelines for Provision of Intensive Care Services (GPICS) 2022 recommends robust and evidence-based clinical guidance surrounding early mobilisation and rehabilitation of patients in Intensive Care.

Safe, standardised working practice is essential for the mobilisation of patients with either oral endotracheal tubes (ETT) or percutaneous tracheostomies with an ETT.

High quality, progressive rehabilitation should be undertaken at the earliest point it is safe and feasible for each individual patient. Excellent communication, teamwork and thorough preparation minimises the risk of harm to the patient.

Not all patients will be appropriate for mobilisation therefore multidisciplinary selection, discussion and agreement is essential.

Inadequate resources may prevent mobilisation on safety grounds.

All Intensive Care staff should be aware of this guideline.

1. INTRODUCTION

Physiotherapists should promote early mobilisation and prevention of deconditioning during periods of acute illness, in addition to providing specialist rehabilitation following critical illness or severe injury (GPICS 2022.)

The intensive care physiotherapy service or, where appropriate, as part of the MDT, should have robust and evidence-based clinical guidance and standard operating procedures surrounding specialist rehabilitation interventions including the early mobilisation/rehabilitation of patients in intensive care (Twose et al, (2019), Hodgson & Tipping (2017), and Sommers et al (2015), GPICS, 2022).

Within the Royal Devon University Healthcare NHS Foundation Trust Northern Services Intensive Care Unit (ICU) mobilisation of patients with endotracheal tubes involves multi-professional working and safety is of our paramount concern.

The aim of mobilisation on the ICU includes:

- improving respiratory function by optimising ventilation/perfusion, increasing lung volumes, and improving secretion clearance,
- reducing the detrimental effects of immobility, e.g., risk to pressure areas and muscle wasting,
- stimulating improved levels of consciousness,
- increasing functional independence,
- improving cardiovascular fitness,
- improving psychological wellbeing, and
- reducing ICU and hospital length of stay.

2. PURPOSE

2.1. This document outlines best practice in the mobilisation of patients with endotracheal tubes.

3. SCOPE

3.1. This guideline relates to the following staff groups who may be involved in mobilising ventilated patients in ICU:

- ICU Nurses
- Physiotherapists
- Physiotherapy Assistants
- Band 4 Nursing Associates
- Health Care Assistants
- Intensive Care Consultants
- Non-consultant ICU doctors

4. DEFINITIONS

- **HCA:** Health Care Assistant
- **ETT:** Endo Tracheal Tube
- **ICU:** Intensive Care Unit
- **GPICS:** Guidelines for the provision of Intensive Care Units.
- **NICE:** National Institute for Health and Care Excellence
- **F_iO_2:** Fractured of inspired oxygen

5. DUTIES AND RESPONSIBILITIES OF STAFF

Roles and responsibilities of all physiotherapists

As part of the multidisciplinary team, physiotherapists must ensure the completion of a comprehensive clinical assessment of those at risk of or with identified physical and non-physical morbidity within four days of admission to Intensive Care and before discharge from Intensive Care (GPICS 2022). This should include the collaborative setting of individualised, patient-centred rehabilitation goals (NICE 2009; 2016)

All patients in the ICU should be screened and considered for daily mobilisation by a physiotherapist within 4 days of admission (GPICS 2022).

Responsibilities of the Advanced Practitioner Respiratory Physiotherapist:

- Ensure access to training for all physiotherapists that participate on the on-call rota.
- Work with the ICU nurse educator to assist with training ICU nurses.
- Promote culture of positivity towards mobilisation in critical care.
- Keep up to date with relevant literature related to safety in the critical care patient. (GPICS 2022)

Intensive Care clinical governance team:

The clinical governance team are responsible for reviewing the document for governance.

Duty Intensive Care consultant:

Has final responsibility in deciding the risk-benefit for the mobilisation of each individual patient. This involves taking into account staffing, skill mix, workload, and factors external to critical care.

Roles and responsibilities of all members of the team:

To have a responsibility to encourage and support mobility of patients as they feel it is safe and appropriate.

For patients who are on sedation, a reduction or hold of sedation should be discussed between the nurses, consultant, and physiotherapist to aim to enable safe mobilisation if appropriate. This will be individual to each patient depending on their medical needs. Where complex cases arise, support will be offered from the ICU consultant who has overall responsibility for the patient's care.

A suitable time for sedation reduction and mobilisation should be agreed upon depending on the severity of risk. If part of the risk assessment is that ICU consultant should be on the unit or immediately available, then mobilisation will not occur until this is achieved.

The physiotherapist and the patient's nurse or nurse in charge must ensure adequate participating staff and skill mix.

The practical plan of action should be discussed and agreed with the team involved.

Designation of roles and clear communication is essential to ensure all persons involved are adequately informed. This is paramount to patient safety and ensures that infusion lines and drains are kept safe.

The team should work together to coordinate this but ultimately the physiotherapist, or team leader, should ensure all aspects of safety are covered. It is essential that one person should be designated to hold the ETT. Other roles may include looking after the central or peripheral lines, or support of the patient from either behind or the side.

Nurses should ensure all non-essential lines are disconnected prior to mobilisation.

Where a patient is very mobile with an ETT a bare minimum of two people should still be in the bed space when the patient is mobilising.

If the plan involves mobilisation to a chair, all staff must be aware of the plan for returning the patient to bed and the proposed timeframe. If for any reason this needs to occur sooner or the physiotherapist is delayed bleep 089; alternatively,327 should be contacted.

Documentation of transfers:
Once the patient has been assessed by the physiotherapist and deemed able to transfer with the nurses more regularly, the number of participants and equipment needed should be communicated to the nurse in the bed space and documented in the notes. This is to aid the physiotherapy and nursing team with risk assessment for future mobility. It's important to include the specific type of mobility aid and the number of people needed for assistance e.g., sara steady standing frame and assistance of 3 to go from bed to chair.

Frequent reassessment and updating of the documentation of the patient's mobilisation dependency should take place by physiotherapists.

Before moving ensure to check:

- Consider consent or if it is in patient's best interest.
- Adequate pain relief, if needed.
- Consider risk-benefit.
- Check for absolute contraindications.
- Considerations around any sedation.
- ETT position at the teeth or lips.

- Anchor Fast/ties are secure (no more than 1 finger under ties).
- Check all points on the ventilator tubing are secure.
- Check airway patent with suction prior to mobilising.
- Assess for and clear any upper airway secretions.
- Physiotherapist has considered readiness: strength, balance, coordination, sensation, etc.
- Nurses disconnect any lines.
- Team to discuss any monitoring needed, change in alarms for movement.
- Check portable ventilator if using.
- Full oxygen CD cylinder if using and 1 spare full cylinder nearby.
- Check portable suction available.
- Clip any drains to patients' gown or ensure safety.
- Ensure dignity of patient.
- Ensure wheelchair or commode is available and nearby in case patient tires quickly.

Designated roles while mobilising include:

- Holding the ETT.
- Instructing the patient, ensuring safety of the patient using the walking aid, and ensuring safe passage.
- Ensure a wheelchair or commode is nearby in case the patient needs to sit down.
- Carry or wheel the ventilator/oxygen cylinder, monitoring, move infusions/drains/catheters.

Considerations:

Specific considerations for patients with an ETT:

- Consideration of grade of intubation (due to risk of unplanned extubation or decannulation). If known difficult intubation, then risk assess for availability of ICU consultant on the unit prior to mobilisation.
- Level of co-operation, agitation, or tube tolerance of patient with reference to displacement of ETT.
- ETT securing method; Anchor Fast, tube ties, length of ventilation tubing, allocated person during treatment.

Patients with trauma with specific considerations:

- Orthopaedic injuries - fractures and fixation, post-operative instructions and weight bearing status need to be reviewed.
- Spinal stability - any collars or braces required to immobilise spine.
- Traumatic brain injury - stable intracerebral pressure (unlikely as would normally go to Derriford).
- Acute spinal cord injury - considerations for abdominal binders and Salisbury protocol.

Walking patients:

Once mobility improves there are two options for mobilising patients:

Walking on the spot at the bedside:

After a thorough physiotherapy assessment, patients can be mobilised on the spot or sidestep up the bed using a hand hold or appropriately assessed aid. It is particularly important to take care when using aids due to the risk of lines or tubing getting caught on them. Therefore, it is important to have one person designated to lines and another specifically always taking care of the ETT.
Physiotherapists who do not work regularly on critical care or who cover ICU as part of their weekend working commitments will be signed off in ICU related competencies covering mobilising patients with ETTs to link to the Association Chartered Physiotherapists In Respiratory Care standards and GPICS 2022 (Skinner et al, (2016), Twose et al (2019). It is not always feasible for these staff to get experience with mobilising patients away from the bed space due to a variety of reasons within their induction period mainly due to working in a small critical care unit and band 5 physiotherapists doing short 4-month rotations. Although every effort is given to enable this by the senior physiotherapy team, this may mean that members of staff may only feel confident to walk the patient on the spot at the bed space rather than away from the bed space.

Walking patient away from bedspace:

If the patient is being mobilised away from the bed space the whole team should be consulted to ensure it is a safe time to do this. Consideration must be made to the consultant's and nurses' workload to ensure they are available in the unlikely event of any emergency. In the case of a dislodged tube the emergency call bell would be pulled to seek help.
When mobilising patients away from the bed space there are 2 options for ventilated patients.

- Leave the patient on their current Dräger ventilator and use a portable CD O2 cylinder. Once you disconnect the power source the Dräger ventilators will only allow you to use an FiO2 1.0 so ensure a full O2 CD cylinder.
- Use one of the two portable ventilators. This involves a disconnection, but oxygen can be titrated specifically. The nurse or physiotherapist must have training to use the portable ventilators and they must be checked prior to any use.

For tracheostomy patients that don't require ventilation, options include:

- Use current ventilator set up with portable O2 and humidification, will default to FiO2 1.0. This allows for change in settings if patient may need increased respiratory pressure assistance while mobilising.
- Use Swedish Nose with oxygen attached via portable full oxygen cylinder.
- Use tracheostomy mask with venturi valve and portable oxygen.
- Use Swedish Nose or Heat Moisture Exchanger (HME) if no oxygen is needed.

For ventilated patients, a minimum of 4 people will be needed when a patient is mobilising away from the bed space to ensure safety. One

person should be allocated to the ETT. All persons involved should have a named task.

Vasopressors

Often patients with endotracheal tubes are receiving vasopressor support. However, patients may be on vasopressors without invasive ventilation.
Physiotherapists should discuss with the nursing staff and the ICU consultant to determine the stability of the patient to determine how appropriate it is to mobilise. Often it is helpful to know how patients are coping with rolling, and the direction of trend in the past hours and the amount of drug being given (Jacob et al 2021.) The stability / trend is more important than the amount of vasopressor, when mobilising out of bed. Up to 0.2 mcg/kg/min of noradrenaline, or equivalent vasopressor seems safe to mobilise (Lindholdz 2022). If greater than 0.2 mcg/kg/min noradrenaline, then the consultant needs to make a risk-benefit assessment and will be responsible. As contraindication the following targets have been accepted in the past. Noradrenaline >0.2 mcg/kg/min for out of bed exercise and > 0.33 mcg/kg/min for in bed exercise (Lindholdz 2022).

For patients mobilising on vasopressors a nurse who can actively manage vasopressors must be available to support physiotherapists, HCAs, and band 4 support staff. The team should anticipate a slightly higher risk of adverse events, mainly transient hypotension requiring a transient escalation in vasopressors (Rebel et al 2019). Being on vasopressors isn't a contraindication to mobilisation and the team should monitor risk considering the clear benefits of mobilisation of these patients (Morais et al 2020; Aneke et al 2019.) A cautious approach with slow progressive mobilisation of patients helps to achieve higher levels of mobility (Jacob et al 2021.)

Blood pressure should be measured for all patients that are on vasopressors prior to mobilisation and be taken again on sitting on the edge of bed. Care needs to be taken of the arterial line pressure transducer. This should be moved with the patient and then repositioned at an approximate level of the patient's heart, at the 4th intercostal space, in the mid-axillary line. It can be feasible to tape or fasten the transducer to the patient's upper arm at an appropriate level. If there are issues with the functioning of the arterial line for blood pressure measurements a manual blood pressure can be taken but it is not a substitute for invasive BP monitoring when continuous vasopressors at a higher level or with frequent adjustments are needed.

6. Literature considerations:

Below shows the most recent evidence from a literature review for consideration when mobilising patients.

Patient readiness should be discussed with the ICU consultant and the patient's nurse. The aim is to ensure safety for all patients and to reduce the risk of adverse events.

Recommendations for contraindications to mobilisation are listed below. If any are present, then a risk-benefit decision needs to be made by the ICU consultant.

Acute cardiac ischemia (Sommers et al 2015, Hodgson 2014, 2015).
Heart rate <40 or >140 (Sommers., et al 2015, Hodgson 2014,2015).
Unless physiologically normal, mean arterial pressure <60 mmHg >110mmHg (Sommers et al 2015, Hodgson 2014, 2015).
O_2 requirement F_iO_2>0.6 with S_a0_2<90% for those with normal targets.
CPAP / NIV dependency requiring a PEEP of more than 10cmH_2O and F_iO_2 >0.6. (Sommers et al 2015, Hodgson 2014, 2015)
A respiratory rate >40 breaths per minute (Sommers., et al 2015).
Noradrenaline >0.2 mcg/kg/min for out of bed exercise and > 0.33 mcg/kg/min for in bed exercise (Lindholdz 2022).
Richmond Agitation Sedation Score -4,-5,+3,+4 (Sommers., et al 2015, Hodson, 2014,2015).
Uncontrolled seizures (Hodson, 2014,2015).
Haemoglobin level <70 (transfuse first).
Known uncontrolled active bleeding (Hodgson., 2014, 2015).
Large open surgical wounds - chest, sternum or abdomen if documented not for mobilisation (Hodgson 2014,2015).
Active management of intracranial pressure that is not in desired range (Hodgson., 2014,2015).
Renal replacement therapy with difficult access.
Neuromuscular blocking drugs.
Uncontrolled pain.
Extreme fatigue.
Pallor and perfuse sweating.
Liver laceration where stipulated they need bed rest by surgical team.
Pulmonary embolus without anticoagulation. To discuss with ICU consultant.
Mobilising in the first 1-2 days post intubation increases the risk of adverse events from 4% to 8% (Hodgson, et al 2022).

7. ARCHIVING ARRANGEMENTS

The original of this SOP, will remain with the author Physiotherapist in Respiratory Care / Advanced Practice Physiotherapist. An electronic copy will be maintained on the Trust intranet, P – Policies – *C – Consent, for example*. Archived electronic copies will be stored on the Trust's “archived policies” shared drive, and will be held indefinitely. A paper copy (where one exists) will be retained for 10 years.

PROCESS FOR MONITORING COMPLIANCE WITH AND EFFECTIVENESS OF THE STANDARD OPERATING PROCEDURE/ GUIDELINE

a. To evidence compliance with this policy, the following elements will be monitored:

What areas need to be monitored?	How will this be evidenced?	Where will this be reported and by whom?
Incidents related to mobility in ICU	*Notes audits* *Datix review group*	*ICU governance*

	Attendance at governance groups	

8. REFERENCES

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APPENDIX 1: COMMUNICATION PLAN

The following action plan will be enacted once the document has gone live.

Staff groups that need to have knowledge of the guideline/SOP	*As per section in sop*
The key changes if a revised document	*na*
The key objectives	*Safety related to early mobility in the intensive care unit.* *Ensure GPICS recommendations are adhered too.*
How new staff will be made aware of the procedure/guideline and manager action	*Physiotherapy:* • *Induction for ICU / on-call* • *On-call rotation* • *On-call document* *Nursing: Induction for ICU* *Step competency's*
Specific Issues to be raised with staff	*As per safety sections within sop*
Training available to staff	*Induction for ICU / on-call and time spent on ICU with clinicians.* *On-call mentor in which staff can highlight training needs and feeds into competency documents.*
Any other requirements	*Na*
Issues following Equality Impact Assessment (if any)	*Na*
Location of hard / electronic copy of the document etc.	*G:\SHARED THERAPY\PHYSIO IP Med_Surg Resp ICU Guidelines*